

et al., 2023b). Further advancements include *causal proxy models* (Wu et al., 2023a), which address the challenge of counterfactual observations.

Applications of causal abstraction span from linguistic phenomena analysis (Arora et al., 2024; Wu et al., 2022b), and evaluation of interpretability methods (Huang et al., 2024), to improving performance through representation finetuning (Wu et al., 2024), and improving efficiency via model distillation (Wu et al., 2022b).

4.3.3 Hypothesis Testing

In addition to the causal abstraction framework, several methods have been developed for rigorous hypothesis testing about neural network behavior. These methods aim to formalize and empirically validate explanations of how neural networks implement specific behaviors.

Causal scrubbing (Chan et al., 2022) formalizes hypotheses as a tuple $(\mathcal{G}, \mathcal{I}, c)$, where $\mathcal{G}$ is the model’s computational graph, $\mathcal{I}$ is an interpretable computational graph hypothesized to explain the behavior, and c maps nodes of $\mathcal{I}$ to nodes of $\mathcal{G}$. This method replaces activations in $\mathcal{G}$ with others that should be equivalent according to the hypothesis, measuring performance on the scrubbed model to validate the hypothesis.

Locally consistent abstractions (Jenner et al., 2023) offer a more permissive approach, checking the consistency between the neural network and the explanation only one step away from the intervention node. This method forms a middle ground between the strictness of full causal abstraction and the flexibility of causal scrubbing.

These methods form a hierarchy of strictness, with full causal abstractions being the most stringent, followed by locally consistent abstractions and causal scrubbing being the most permissive. This hierarchy highlights trade-offs in choosing stricter or more permissive notions, affecting the ability to find acceptable explanations, generalization, and mechanistic anomaly detection.

4.4 Integrating Observation and Intervention.

To comprehensively understand internal neural network mechanisms, combining observational and interventional methods is crucial. For instance, sparse autoencoders can be used to disentangle superposed features (Cunningham et al., 2024), followed by targeted activation patching to test the causal importance of these features (Wang et al., 2023). Similarly, the logit lens can track prediction formation across layers (nostalgabraist, 2020), with subsequent interventions confirming causal relationships at key points. Probing techniques can identify encoded information (Belinkov, 2021), which can then be subjected to causal abstraction (Geiger et al., 2023a) to understand how this information is utilized. This iterative refinement process, where broad observational methods guide targeted interventions and intervention results inform further observations, enables a multi-level analysis that builds a holistic understanding across different levels of abstraction. Recent work (Marks et al., 2024; Bushnaq et al., 2024; Braun et al., 2024; O’Neill & Bui, 2024; Ge et al., 2024) demonstrates the potential of integrating sparse autoencoders with automated circuits discovery (Conmy et al., 2023; Syed et al., 2023), combining feature-level analysis with circuit-level interventions to uncover the interplay between representation and mechanism.

5 Current Research

This section surveys current research in mechanistic interpretability across three approaches based on when and how the model is interpreted during training: Intrinsic interpretability methods are applied before training to enhance the model’s inherent interpretability (Section 5.1). Developmental interpretability involves studying the model’s learning dynamics and the emergence of internal structures during training (Section 5.2). After training, post-hoc interpretability techniques are applied to gain insights into the model’s behavior and decision-making processes (Section 5.3), including efforts towards uncovering general, transferable principles across models and tasks, as well as automating the discovery and interpretation of critical circuits in trained models (Section 5.4).



Figure 10: Key desiderata for interpretability approaches across training and analysis stages: (1) Intrinsic: Architectural biases for sparsity, *modularity*, and *disentangled* representations. (2) Developmental: Predictive capability for phase transitions, manageable number of critical transitions, and a unifying theory connecting observations to singularity geometry. (3) Post-hoc: Global, comprehensive, automated discovery of critical circuits, uncovering transferable principles across models/tasks, and extracting high-level causal mechanisms.

5.1 Intrinsic Interpretability

Intrinsic methods for mechanistic interpretability offer a promising approach to designing neural networks that are more amenable to *reverse engineering* without sacrificing performance. By encouraging *sparsity*, *modularity*, and *monosemanticity* through architectural choices and training procedures, these methods aim to make the reverse engineering process more tractable.

Intrinsic interpretability methods aim to constrain the training process to make learned programs more interpretable (Friedman et al., 2023b). This approach is closely related to neurosymbolic learning (Riegel et al., 2020) and can involve techniques like regularization with spatial structure, akin to the organization of information in the human brain (Liu et al., 2023a;b).

Recent work has explored various architectural choices and training procedures to improve the interpretability of neural networks. Jermyn et al. (2022) and Elhage et al. (2022a) demonstrate that architectural choices can affect monosemanticity, suggesting that models could be engineered to be more *monosemantic*. Sharkey (2023) propose using a bilinear layer instead of a linear layer to encourage monosemanticity in language models.

Liu et al. (2023a) and Liu et al. (2023b) introduce a biologically inspired spatial regularization regime called brain-inspired modular training for forming modules in networks during training. They showcase how this can help RNNs exhibit brain-like anatomical modularity without degrading performance, in contrast to naive attempts to use sparsity to reduce the cost of having more neurons per layer (Jermyn et al., 2022; Bricken et al., 2023).

Preceding the mechanistic interpretability literature, various works have explored techniques to improve interpretability, such as sparse attention (Zhang et al., 2021), adding L^1 penalties to neuron activations (Kasioumis et al., 2021; Georgiadis, 2019), and pruning neurons (Frankle & Carbin, 2019). These techniques have been shown to encourage sparsity, modularity, and disentanglement, which are essential aspects of intrinsic interpretability.

5.2 Developmental Interpretability

Developmental interpretability examines the learning dynamics and emergence of internal structures in neural networks over time, focusing on the formation of *features* and *circuits*. This approach complements static analyses by investigating critical phase transitions corresponding to significant changes in model behavior or capabilities (Steinhardt, 2023; Schaeffer et al., 2023; Wei et al., 2022; Simon et al., 2023). While primarily a distinct field, developmental interpretability often intersects with mechanistic interpretability, as exemplified by Olsson et al. (2022)’s work. Their research, rooted in mechanistic interpretability, demonstrated how

the emergence of in-context learning relates to specific training phase transitions, connecting microscopic changes (induction heads) with macroscopic observables (training loss).

A key motivation for developmental interpretability is investigating the *universality* of safety-critical patterns, aiming to understand how deeply ingrained and thereby resistant to safety fine-tuning capabilities like deception are. In addition, researchers hypothesize that emergent capabilities correspond to sudden circuit formation during training (Michaud et al., 2023), potentially allowing for prediction or control of their development.

Singular Learning Theory (SLT), developed by Watanabe (Watanabe, 2009; 2018), provides a rigorous framework for understanding overparameterized models’ behavior and generalization. By quantifying model complexity through the *local learning coefficient*, SLT offers insights into learning phase transitions and the emergence of structure in the model (Lau et al., 2023). Recent work by Hoogland et al. (2024) applied this coefficient to identify developmental stages in transformer models, while Furman & Lau (2024) and Chen et al. (2023b) advanced SLT’s scalability and application to the toy model of *superposition* (Figure 4), respectively.

While direct applications to phenomena such as generalization (Zhang et al., 2017), learning functions with increasing complexity (Nakkiran et al., 2019), and the transition from memorization to generalization (*grokking*) (Liu et al., 2022a; Power et al., 2022; Liu et al., 2022b; Nanda et al., 2023a; Varma et al., 2023; Thilak et al., 2022; Merrill et al., 2023; Liu et al., 2023c; Stander et al., 2023; Wang et al., 2024) are limited, these areas, along with neural scaling laws (Caballero et al., 2022; Liu & Tegmark, 2023; Michaud et al., 2023) (which can be connected to mechanistic insights (Hernandez et al., 2022)), represent promising future research directions.

In conclusion, developmental interpretability serves as an evolutionary theory lens for neural networks, offering insights into the emergence of structures and behaviors over time (Saphra, 2023). Drawing parallels from systems biology (Alon, 2019), this approach can apply concepts like network *motifs*, robustness, and *modularity* to neural network development, explaining how functional capabilities arise. Sometimes, understanding how structures came about is easier than analyzing the final product, similar to how biologists find certain features in organisms easier to explain in light of their evolutionary history. By studying the temporal aspects of neural network training, researchers can potentially uncover fundamental principles of learning and representation that may not be apparent from examining static, trained models alone.

5.3 Post-Hoc Interpretability

In applied mechanistic interpretability, researchers explore various facets and methodologies to uncover the inner workings of AI models. Some key distinctions are drawn between *global* versus *local* interpretability and *comprehensive* versus *partial* interpretability. Global interpretability aims to uncover general patterns and behaviors of a model, providing insights that apply broadly across many instances (Doshi-Velez & Kim, 2017; Nanda, 2023e). In contrast, local interpretability explains the reasons behind a model’s decisions for particular instances, offering insights into individual predictions or behaviors. Comprehensive interpretability involves achieving a deep and exhaustive understanding of a model’s behavior, providing a holistic view of its inner workings (Nanda, 2023e). In contrast, partial interpretability often applied to larger and more complex models, concentrates on interpreting specific aspects or subsets of the model’s behavior, focusing on the application’s most relevant or critical areas.

Large Models – Narrow Behavior. Circuit-style mechanistic interpretability aims to explain neural networks by *reverse engineering* the underlying mechanisms at the level of individual neurons or subgraphs. This approach assumes that neural vector representations encode high-level concepts and circuits defined by model weights encode meaningful algorithms (Olah et al., 2020; Cammarata et al., 2020). Studies on deep networks support these claims, identifying circuits responsible for detecting curved lines or object orientation (Cammarata et al., 2020; 2021; Voss et al., 2021).

This paradigm has been applied to language models to discover subnetworks (circuits) responsible for specific capabilities. Circuit analysis localizes and understands subgraphs within a model’s computational graph responsible for specific behaviors. For large language models, this often involves narrow investigations into